Table 2: Key advantages and benefits of using cloud SDR

Advantage	Practical impact	Technical benefit
Remote web access	No hardware required	Browser-based control
Global receiver network	Propagation studies	Geographic diversity
Wide HF coverage	Shortwave monitoring	0–30 MHz support
Cross-platform access	Universal usability	OS independent
Secure remote login	Controlled usage	Access control options
No installation needed	Quick deployment	Cloud-hosted software
Multi-user capability	Collaborative research	Shared infrastructure
Centralised data logging	Long-term analysis	Cloud storage support
Low infrastructure cost	Budget-friendly access	Shared hardware model
TDoA geolocation	Signal source tracking	Distributed receivers
GPS synchronisation	Precise measurements	Accurate timing reference
Public accessibility	Learning opportunities	Open network nodes
Scalability	Wide-area monitoring	Add remote nodes
Reduced maintenance	Minimal user effort	Server-side updates
Real-time waterfall	Instant signal detection	Live spectrum view
AI integration potential	Automated classification	Cloud analytics engines
Disaster resilience	Service continuity	Distributed architecture
Virtual lab capability	Academic training support	Remote experimentation

algorithms, and machine learning integration. With the integration of AI and data analytics, SDR systems are now quite capable of working with automated signal classification, threat detection, illegal use of radio frequencies, anomaly detection and cognitive radio experimentation.

SDR on cloud for analysis of radio frequencies

SDR's capabilities are not limited to local deployment of connected hardware. Cloud-based SDR systems are now providing users the access to remote radio receivers deployed in different geographic locations. A popular and high-performance platform is KiwiSDR (*http://kiwisdr.com/*), which enables remote users to tune into shortwave and other frequencies through a web browser interface. It provides a cloud-based environment for linking with different radio frequencies and bands at different locations in the world.

Figure 2: KiwiSDR as cloud SDR for radio spectrum analysis and intelligence